Scientific Computation Project 2

Your college ID here

December 9, 2024

Part 1

1.

(a-1) This is a nonlinear system of equations. For the non-trivial equilibirum, we need to find solutions numerically using the root function in scipy.optimize. The default (hybr) and Krylov methods are used to check the correctness. The system of equations are specified using the provided model code after removing the t parameter. A "guess" for the solution is also given. Then the iterative approach moves from the guess towards a solution one step at a time.

(a-2) Once the equilibrium is found, we linearize the system about this equilibrium to get the Jacobian matrix J. We then compute the matrix exponential e^{JT}, and find the initial perturbation vector $\tilde{x}_0$ that leads to the largest energy growth ratio $\frac{e(T)}{e(0)}$. This is done by computing the singular value decomposition (SVD) of e^{JT} and picking the singular vector associated with the largest singular value.

(b) We test consistency by taking the initial perturbation vector $\tilde{x}_0$ found in (a), simulating the full nonlinear system from $x = \bar{x} + \tilde{x}_0$, and then comparing the observed ratio $e(T)/e(0)$ from the nonlinear simulation with the linearized prediction. The outputs, "Linear prediction of max e(T)/e(0) in part1q1a: 26879.887079694287, Nonlinear simulation e(T)/e(0) in simulate1: 26870.19029856483", shows the consistency between these two methods.

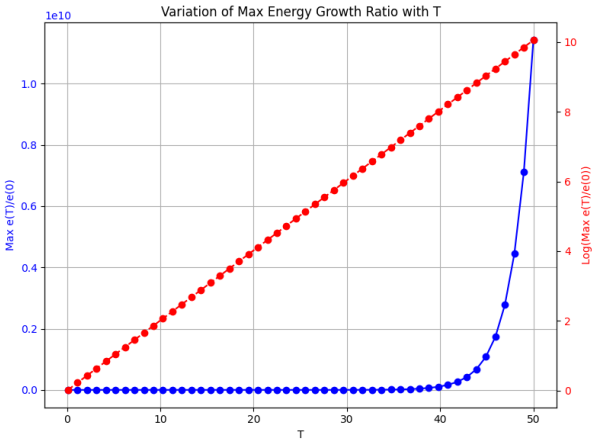


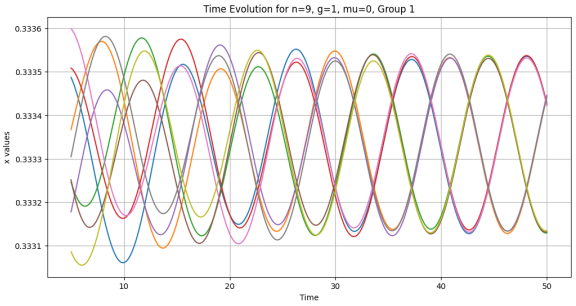
Figure 1: Exponential growth with increasing timestep

(c) The system exhibits exponential instability over time, with perturbations growing significantly as T increases. $\gamma = 2$ amplifies coupling effects between system components and promotes instability via increasing the magnitude of unstable eigenvalues in J. With $\mu = 0$, stabilizing terms in the system are removed, making it easier for instability to dominate.

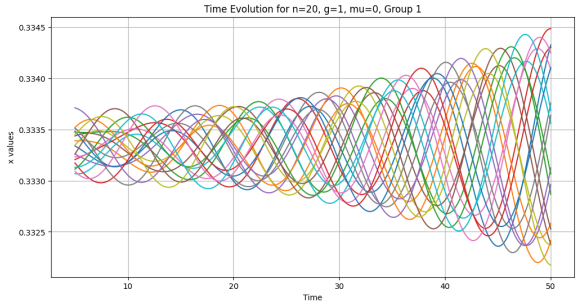
2.

(case A: n = 9) The long-term dynamics are periodic and well-ordered, with no evidence of chaos. The system appears to settle into stable sinusoidal oscillations. The global qualitative dynamics are analysed below:

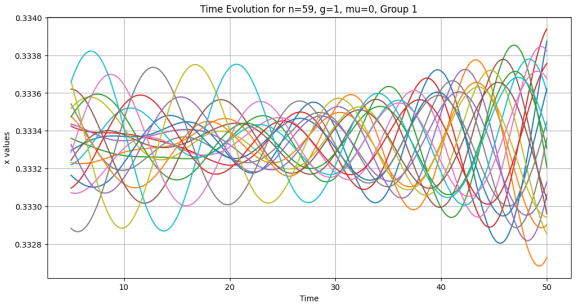
- Time-dependence: Yes. The trajectories clearly exhibit time-dependent behavior, with oscillations occurring over time.
- Simple sinusoidal oscillations: Yes, the trajectories exhibit simple sinusoidal oscillations. Each component x_i appears to follow a regular and periodic sinusoidal pattern.
- Synchronized components: Yes, the components are synchronized. All trajectories maintain a coherent phase and amplitude relationship, indicating strong coupling between the components.
- Chaotic dynamics: No chaotic dynamics are present. The regular and sharply decaying PSD is characteristic of periodic or quasi-periodic motion.



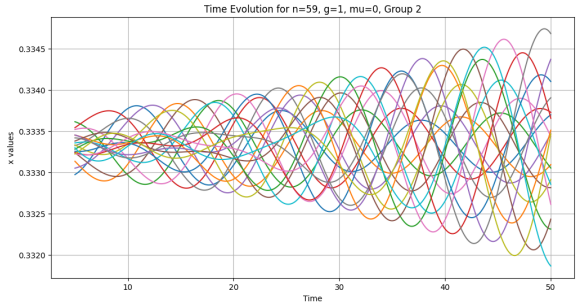
(a) case A: $n = 9$



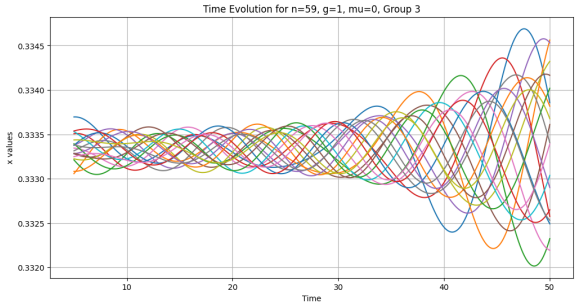
(b) case B: $n = 20$



(c) Group 1 with first 20 components



(d) Group 2 with second 20 components



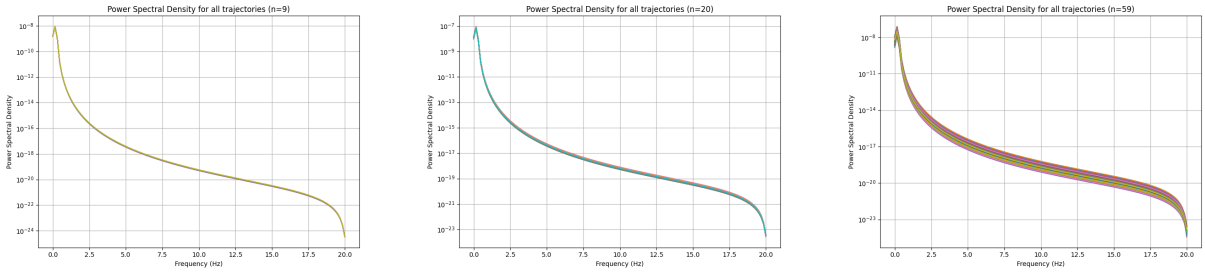
(e) Group 3 with last 19 components

Figure 2: Simulation results for three cases with $\gamma = 1$ and $\mu = 0$

(case B: n = 20) The global qualitative dynamics are analysed below:

- Time-dependence: Yes, the system is time-dependent, as evident from the oscillatory trajectories.
- Simple sinusoidal oscillations: the trajectories show sinusoidal oscillations, but the oscillations are less uniform than in Case A (n=9). There is noticeable variation in amplitude and phase across the components.
- Synchronized components: Partial synchronization is observed. While the components maintain similar frequencies, there are phase shifts and amplitude differences among them, reducing the overall coherence compared to Case A (n=9).

- Chaotic dynamics: No evidence of chaotic dynamics is present. The PSD still reflects stable periodic or quasi-periodic motion, although some trajectories exhibit slight deviations in amplitude.



(a) case A: $n = 9$ (b) case B: $n = 20$ (c) case C: $n = 59$

Figure 3: Power Spectral Density (PSD) results for three cases with $\gamma = 1$ and $\mu = 0$

(case C: n = 59) The oscillation amplitudes increase compared to Cases A and B, reflecting the greater degrees of freedom and interactions in the larger system.

- Time-dependence: Yes, oscillatory trajectories shows the system is time-dependent.
- Simple sinusoidal oscillations: No, sinusoidal oscillations are observed, but with more variability in amplitude and phase compared to Cases A and B. The oscillations are not uniform, and some trajectories exhibit larger deviations.
- Synchronized components: No. The trajectories show noticeable phase differences and amplitude variations, indicating a weakening of global coherence as n increases.
- Chaotic dynamics: The system exhibits more complex, quasi-periodic behavior with reduced synchronization, but it does not show strong evidence of chaotic dynamics. The PSD does not exhibit the hallmark features of chaos (e.g., broad, continuous spectrum over a wide frequency range or high power at intermediate and high frequencies).

Part 2

1.

Efficient reasons in matrix memory and solver: The matrix in the dualfd1 function is dense with the shape $2n \times 2n$, while the matrix in the part2q1 function is sparse with the shape $7 \times 2n$, reducing excessive memory due to storing the zeros in A and reducing excessive operations due to the zeros. During our working process, we tried the spslove version which solves the memory issue, but finally we chose the solve_banded which is specially designed for banded matrices with actually using a Fortan routine from the Lapack library. During our process, we use the inefficient dualfd1 to check the correctness of our method due to the solve_banded requires a tricky matrix input.

2. multiscale problem

The test function is $u(x, y) = sin(k_1 x) + sin(k_2 y)$ where $k_1 = 8 * \pi/5$ and $k_2 = 2 * \pi$ represent the frequencies. The Grid size ranges from 10 to 400, as m is comparable to n, so keep m = n for simplicity.

[Time] Implicit schemes are computationally expensive (as seen in the time comparison), especially for large n, because they require solving a system of equations; explicit schemes are computationally cheaper. The running time of implicit part2q1 increases exponentially while the explicit fd2 is relatively low as the grid size n increases.

[Accuracy] When n doubles, the error reduces by 1/4. For example, n increases from 100 to 200, h is halved ($h \propto \frac{1}{n}$) because h = 1 / (n - 1). Since the error scales as h^2, halving h reduces the error by a factor of $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

Initially, part2q1 shows higher errors, which decreases rapidly and then stabilizes, while fd2 starts with lower errors. As the grid resolution increases, the explicit and implicit methods converge to similar accuracies which means that both approaches eventually produce the same errors.

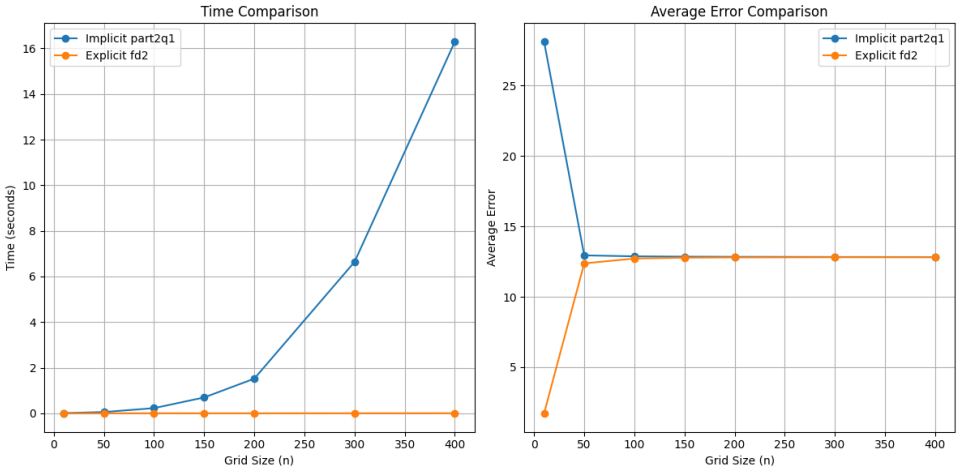
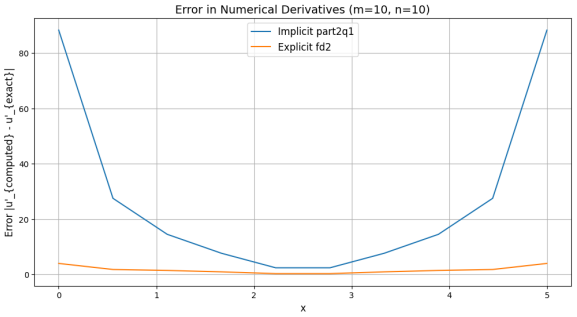
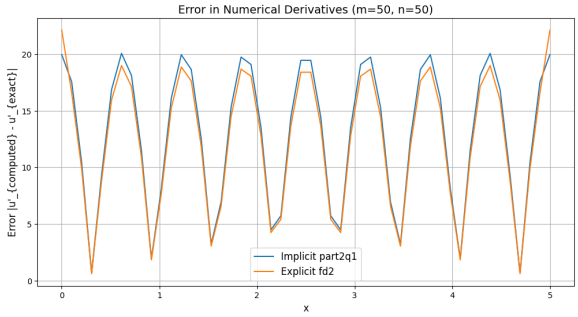


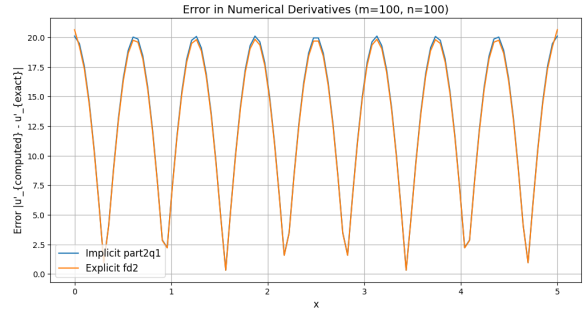
Figure 4: Running time and errors of implicit part2q1 and explicit fd2



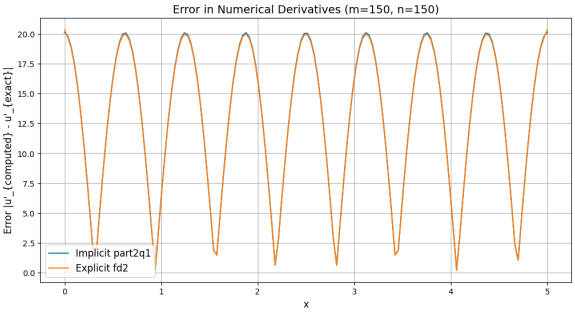
(a) $n = 10$



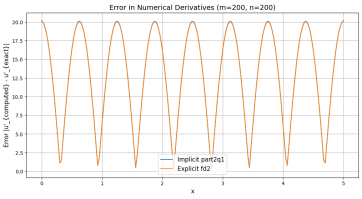
(b) $n = 50$



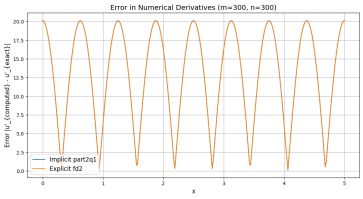
(c) $n = 100$



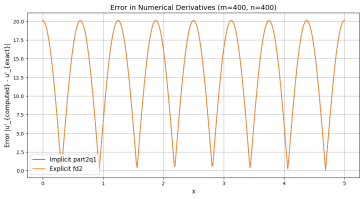
(d) $n = 150$



(e) $n = 200$



(f) $n = 300$



(g) $n = 400$

Figure 5: $|df/dx_{\text{computed}} - df/dx_{\text{exact}}|$ errors of implicit part2q1 and explicit fd2 with different n